

THE PHYSICAL AI DUAL-BRAIN ARCHITECTURE

Proposal-Permission Privilege Split on Heterogeneous Silicon

UNTRUSTED PROPOSAL ENGINE (Linux MPU / Edge NPU)

- Multi-Modal Foundation Models (VLMs), Vision Encoders (ViTs / DINOv2), Action Chunk Decoders (Diffusion / ACT)
- Asynchronous Deliberation: 1 Hz Semantic Intent $\rightarrow$ 20–50 Hz Action Chunking (Amortizes Inference Delay)
- **Emits: Expiring Intent Leases $pt = \langle SE(3) \text{ Target, Workspace Bounding Volume, } t_{\text{expire}}, \text{ Monotonic Counter} \rangle$**

Shared SRAM Mailbox · Expiring Proposal pt (TTL ≤ 100 ms)

TRUSTED PERMISSION AUTHORITY (Real-Time Bare-Metal MCU)

- Dedicated 1000 Hz Timing Loop: Zero Dynamic Memory Heap (Static SRAM buffers only)
- Minimal-Intervention Control Barrier Functions (CBF: $h(x) \geq 0$) + Dynamic Stopping Clearance ($d_{\text{stop}} \leq d_{\text{clear}}$)
- **Two-Stage Emergency Interlock (IEC 60204-1): Heartbeat Watchdog Timeout (> 20 ms) $\rightarrow$ SS1 Dynamic Halt $\rightarrow$ Safe Torque Off**

Permitted Phase Currents $ut = \text{permit}(pt)$ (Space Vector PWM @ 20 kHz)

THE PHYSICAL WORLD (Irreversible State Mutation: $W_t \longrightarrow W_{t+1}$)

- 3-Phase MOSFET Inverters · Stator Magnetic Flux · Kinetic Momentum ($p = mv$) · Joule Heating (I^2R)
- *Endogenous Feedback: Physical Mutation W_{t+1} Instantly Shapes Next Sensory Observation O_{t+1}*